

Conceptual Summary

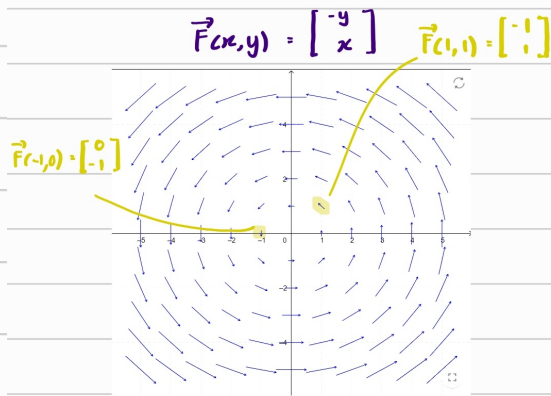
Disclaimer :

- ① Boxes highted in Red must be learned
- ② Example Questions won't be provided here, take a look a "Filtered topical Questions" or topical notes for Example Questions.

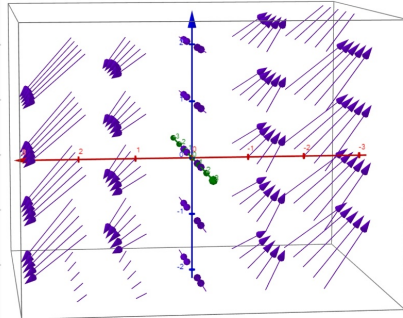
Vector & Del Operator

Vector field

Function that assigns a vector every point in a region.



$$\vec{F}(x,y,z) = \begin{bmatrix} 2x \\ -2y \\ -2z \end{bmatrix} = 2x \underline{i} - 2y \underline{j} - 2z \underline{k}$$



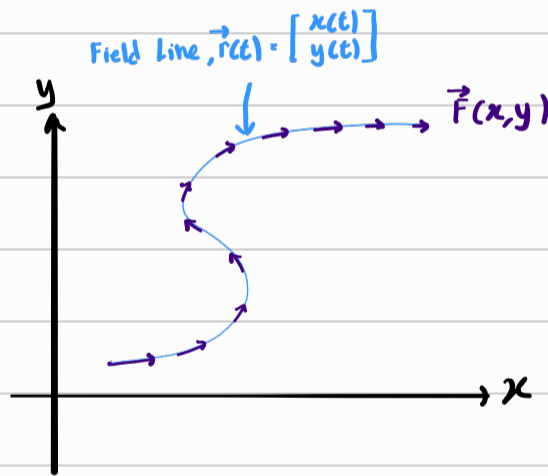
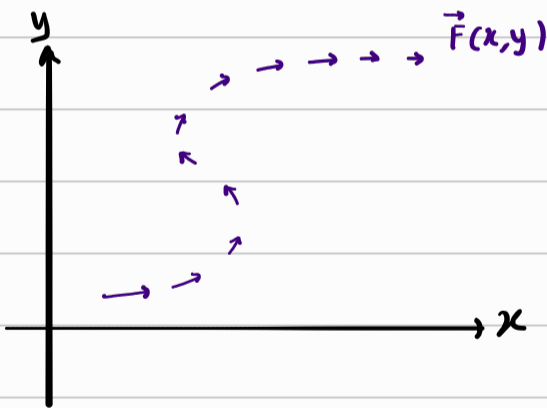
Field Lines

A field line is a smooth curve such that, at every point on the curve, the $\vec{r}'(t)$ tangent direc. agrees w the $\vec{F}(x,y)$ direc. of vector field there.

Notations

$\vec{r}(t) \Rightarrow$ position vector [A point]

$\vec{F}(x,y) \Rightarrow$ Takes a point, returns corresponding vector.



Assuming $\vec{r}(t)$ is a field line to vector field, $\vec{F}(x,y)$, then $\vec{r}'(t)$ is tangent to the field line & $\vec{F}(\vec{r}(t))$ is the field vector at the same point. Since a field line always follows the field direc.

$$\vec{r}'(t) = \vec{F}(\vec{r}(t))$$

Del Operator, ∇

∇ is a vector differential operator

$$\nabla = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix}$$

Here are the following use cases for the del operator

scalar field

① Gradient Vector, ∇f

$$\nabla f(x, y, z) = \frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k}$$

$$\nabla f(r, \theta, z) = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta + \frac{\partial f}{\partial z} \hat{e}_z$$

$$\nabla f(r, \theta, \phi) = \frac{\partial f}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{e}_\phi$$

■ basis vectors

■ Normalising factors

• ∇f collects the gradient in the axis dircc. for each coordinate

vector field / func.

② Divergence, $\nabla \cdot \mathbf{F}$

$$\mathbf{F}(x, y, z) = \begin{bmatrix} F_1(x, y, z) \\ F_2(x, y, z) \\ F_3(x, y, z) \end{bmatrix}$$

$$\nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

Sums up the local stretching rates in x, y, z dircc.

e.g. $\partial F_1 / \partial x > 0$, this means right side pulls away faster, left side fills.

• Divergence measures the local tendency of the field at a point

to spread out from or converge toward that point.

$\therefore$ If $\nabla \cdot \mathbf{F} > 0 \Rightarrow$ Local expansion [source] $\left. \begin{array}{l} \nabla \cdot \mathbf{F} \neq 0 \\ \nabla \cdot \mathbf{F} < 0 \Rightarrow \text{Local compression [sink]} \end{array} \right\} \Rightarrow \text{Compressible}$

$\nabla \cdot \mathbf{F} = 0 \Rightarrow$ No volume change $\Rightarrow$ Incompressible

③ Curl of vector field, $\nabla \times F$

$$F(x,y,z) = F_x \underline{i} + F_y \underline{j} + F_z \underline{k} = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \langle F_x, F_y, F_z \rangle$$
$$\nabla \times F = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \times \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = \begin{bmatrix} \partial F_z/\partial y - \partial F_y/\partial z \\ \partial F_x/\partial z - \partial F_z/\partial x \\ \partial F_y/\partial x - \partial F_x/\partial y \end{bmatrix}$$

Take z -component as an example: $\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}$

• If $\frac{\partial F_y}{\partial x} > 0$, upward "force" to the right is stronger, applying a counter CW torque

• If $\frac{\partial F_x}{\partial y} > 0$, right side "force" to the top is stronger, applying a CW torque

$\therefore$ If $\left| \frac{\partial F_y}{\partial x} \right| = \left| \frac{\partial F_x}{\partial y} \right|$, $\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y} = 0$, no rotation will occur.



Laplacian Operator, ∇^2

The Laplacian measures the [divergence of the gradient] or [net local curvature] of a scalar field.

In other words, how the value at a point compares with the avg of nearby values.

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Integration Stuff

Let's start by understanding what is integration.

$$\int_{x_1}^{x_2} f(x) dx$$

This means summing up each $[f(x) \cdot dx]$ from $x=a$ to $x=b$.

The quantity this integral calculates depends on what $f(x)$ represents.

↳ If $f(x) = \text{height} \Rightarrow \int_{x_1}^{x_2} f(x) dx = \text{Area under } f(x)$

↳ If $f(x) = \text{Density} \Rightarrow \int_{x_1}^{x_2} f(x) dx = \text{Mass of 1D line [e.g. 1D rod]}$

⋮

Double Integral

$$\int_{y_1}^{y_2} \left[\int_{x_1}^{x_2} f(x,y) dx \right] dy$$

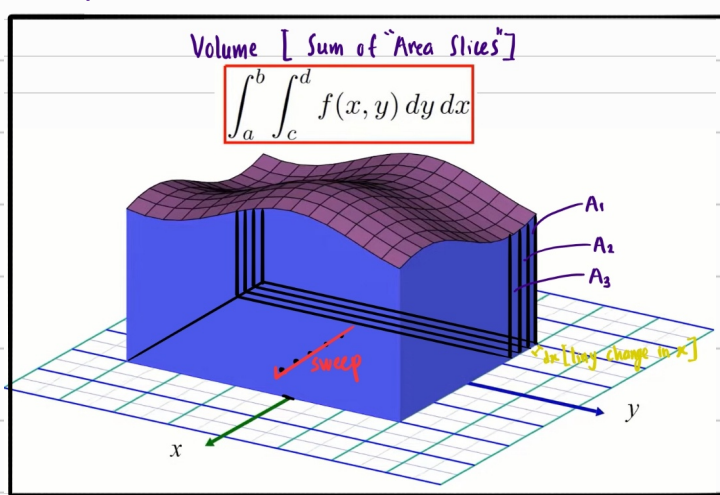
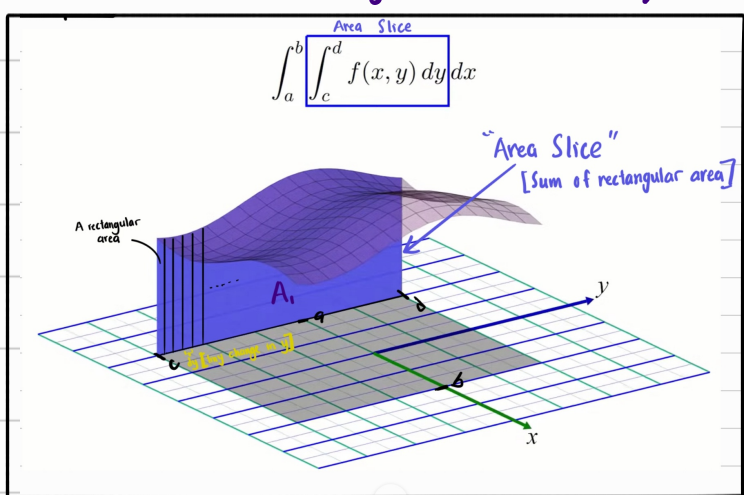
Let's say $f(x,y) = \text{Height} \Rightarrow \text{Area, } A = \int_{x_1}^{x_2} f(x,y) dx$

This gets us an eqn that represents each "Area Slice" whatever x values within x_1 and x_2 .

Then we sum up each of these area slices in the y -direc, y_1 to y_2

$$\int_{y_1}^{y_2} A dy$$

This will then get us an eqn that represents the Volume below $f(x,y)$.



Do notice that in this case, we first got the eqn for the Area slice in the y -direction, then we only sum up each area slice in the x -direction. But it will work vice versa too.

This is called Change in Order

$$\int_{x_1}^{x_2} \left[\int_{y_1}^{y_2} f(x,y) dy \right] dx = \int_{y_3}^{y_4} \left[\int_{x_3}^{x_4} f(x,y) dx \right] dy$$

[Do note that the corresponding boundaries might change]

For more complicated shapes, the integral might get too complicated.
In that case, we can use another coordinate system
[e.g. 2D: Polar, 3D: Spherical or Cylindrical]

Let's say we want to convert this to polar coordinates

$$\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x,y) dx dy$$

We essentially need to change 3 things

- ① Boundaries: x_1, x_2, y_1, y_2 [By looking at graph]
- ② $f(x,y)$ to $f(r,\theta)$ [By substituting $x = r\cos\theta, y = r\sin\theta$]
- ③ Write $dx dy$ in terms of $dr d\theta$

↳ Take note that mostly $dx dy \neq dr d\theta$

↳ To make them equal, a scaling factor is multiplied with $dr d\theta$
that is called Jacobian, $J(r,\theta)$

$$\rightarrow dx dy = J(r,\theta) \cdot dr d\theta$$

$$J(r,\theta) = \det \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}$$

This whole explanation was on the assumption $f(x) = \text{height}$.

What if $f(x,y) = \text{Density}$?

$$\int_{y_1}^{y_2} \left[\int_{x_1}^{x_2} f(x,y) dx \right] dy$$

$\int_{x_1}^{x_2} f(x,y) dx$ = Eqn of mass along x -direc. from x_1 to x_2

or in other words "mass of a line along x axis"

$\therefore \int_{y_1}^{y_2} \left[\int_{x_1}^{x_2} f(x,y) dx \right] dy$ = Sum of "mass of a line along x -axis" from y_1 to y_2
or in other words "mass of an area"

What if we wanted to get "mass of a Volume"? That's where
triple integral comes in.

Tripple Integral

$$\int_{z_1}^{z_2} \left[\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x,y,z) dx dy \right] dz$$

If $f(x,y,z) = \text{Density}$

$$\implies \int_{y_1}^{y_2} \left[\int_{x_1}^{x_2} f(x,y,z) dx \right] dy = \text{Mass of an Area}$$

Then we sum up "Mass of an Area" across the z -axis from z_1 to z_2

$$\int_{z_1}^{z_2} (\text{Mass of area slice}) dz$$

This will then get us the mass of the volume below $f(x,y,z)$

What if $f(x,y,z) = 1$?

Then $\int_{z_1}^{z_2} \left[\int_{y_1}^{y_2} \int_{x_1}^{x_2} f(x,y,z) dx dy \right] dz$ just gets u
the mass of a Volume with density = 1, aka. pure volume.

The explanation for change of order is the same w double integral,
just with 1 extra variable

The explanation for change of coordinate is the same w double integral,
with again 3 things to change.

① Boundaries : $x_1, x_2, y_1, y_2, z_1, z_2$ [By graph]

② $f(x,y,z)$ [By substitution]

③ Write $dx dy dz$ in terms of new coordinate system [e.g. $dr d\theta d\phi$]

$$J(u,v,w) = \det \begin{bmatrix} \partial x / \partial u & \partial x / \partial v & \partial x / \partial w \\ \partial y / \partial u & \partial y / \partial v & \partial y / \partial w \\ \partial z / \partial u & \partial z / \partial v & \partial z / \partial w \end{bmatrix}$$

Line (Curve) Integrals

$$\int f \, ds$$

Line integral is an integral used to accumulate a quantity along a curve.

Previously we only know how to accumulate a quantity along a straight line using

$$\int_{x_1}^{x_2} f(x) \, dx$$

We were essentially summing up each corresponding $f(x)$ of each tiny straight line, dx .

What if we want to accumulate a quantity along a curve?

It's simple, we just sum up each corresponding $f(x,y)$ of each tiny curved line ds .

$$\int f \, ds$$

However, notice that $\int$ can only handle 1 variable, but we might have > 1 variable

To solve this issue, we use a parameterization func., $\vec{r}(t)$

to map every 1D coordinate [In the t coordinate system] to its specific original location [In the original coordinate system]

Again, there have to change 3 things

① Boundaries

② $f(x,y,z)$ to $f(\vec{r}(t))$ [To find quantity on curve only]

③ Write ds in terms of dt

$$\hookrightarrow ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} \, dt$$

$$ds = \left| \frac{d\vec{r}}{dt} \right| \, dt$$

$$\int_{t_1}^{t_2} f(\vec{r}(t)) \left| \frac{d\vec{r}}{dt} \right| \, dt$$

Work Done along curved path

If $f(x, y, z)$ represented the force along the curve

$$\Rightarrow f(x, y, z) = \vec{F} \cdot \vec{T} \quad \text{Unit tangent vector}$$
$$= \vec{F} \cdot \frac{d\vec{r}}{ds}$$

Then the sum of $f(x, y, z)$ will get us the work done along that curve

$$\text{Work done} = \int \left[\vec{F} \cdot \frac{d\vec{r}}{ds} \right] ds$$
$$= \int \left[\vec{F} \cdot \frac{d\vec{r}}{ds} \right] \left(\frac{ds}{dt} dt \right)$$

$$\text{Work done} = \int_{t_1}^{t_2} \left[\vec{F} \cdot \frac{d\vec{r}}{dt} \right] dt$$

Conservative Vector Fields

Vector field whose line integral depends only on the start & end points, not on the path taken.

This is only possible if there are fixed values attached to each point in space, allowing us to ignore the path taken.

$\Rightarrow$ A conservative Vector field must have an underlying function.

$\therefore$ We can say

$$\vec{F} = \nabla f$$

Potential function

$$\int_{t_1}^{t_2} \vec{F} \cdot \frac{d\vec{r}}{dt} dt = \int_{t_1}^{t_2} \nabla f \cdot \frac{d\vec{r}}{dt} dt$$
$$= f(t_2) - f(t_1)$$

$$\text{Properties : } \nabla \times \vec{F} = 0$$

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

Surface Integrals

$$\iint f(x,y,z) dS$$

Up till now, the only surface area formula we know is

$$\iint f(x,y) dx dy, \text{ when } f(x,y) = 1$$

This formula entirely confined to a flat 2D planes since it only has 2 degrees of freedom $[x, y]$.

Or in other words, $dx dy$ is a flat 2D square. Summing up flat 2D squares will only get you flat 2D area

What if we want to find the surface area of a curved surface in 3D?

It's simple, rather than summing up flat 2D squares $[dA = dx dy]$,

we sum up curved 3D squares, dS

$$\iint f(x,y,z) dS$$

However, notice that $\iint$ can only handle 2 variables, but we have 3. (x,y,z)

To solve this issue, we use a parameterization func., $\vec{r}(u,v)$

to map every flat coordinate [In the u,v coordinate system] to its specific

3D location [In the x,y,z coordinate system]

Again, there have to change 3 things

① Boundaries

② $f(x,y,z)$ to $\vec{r}(u,v)$

③ Write dS in terms of $du dv$

$$\hookrightarrow dS = \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv$$

$$\text{Surface Area} = \int_v^{v_2} \int_u^{u_2} \vec{r}(u,v) \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv, \text{ if } f(x,y,z) = 1$$

Flux

Flux is the ammount of vector field going perpendicularly through a surface.

To get this, we just set $f(x,y,z) = \vec{F} \cdot \vec{n}$

Vector Fieldunit normal vector

$$\therefore \text{Flux} = \iint [\vec{F}(x,y,z) \cdot \vec{n}] dS$$

Same issue as previously, so we use a parameterization func., $\vec{r}(u,v)$ and change the 3 things again

① Boundaries

② $\vec{F}(x,y,z) \rightarrow \vec{F}(\vec{r}(u,v))$ [To find vector field at surface only]

③ $dS = \left| \frac{d\vec{r}}{du} \times \frac{d\vec{r}}{dv} \right| du dv = |\vec{r}_u \times \vec{r}_v| du dv$

$$\begin{aligned} \text{Flux} &= \int_{v_1}^{v_2} \int_{u_1}^{u_2} [\vec{F}(\vec{r}(u,v)) \cdot \vec{n}] |\vec{r}_u \times \vec{r}_v| \\ &= \int_{v_1}^{v_2} \int_{u_1}^{u_2} \vec{F}(\vec{r}(u,v)) \cdot \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|} |\vec{r}_u \times \vec{r}_v| \\ &= \int_{v_1}^{v_2} \int_{u_1}^{u_2} \vec{F}(\vec{r}(u,v)) \cdot (\vec{r}_u \times \vec{r}_v) \end{aligned}$$

$$\text{Flux} = \int_{v_1}^{v_2} \int_{u_1}^{u_2} \vec{F}(\vec{r}(u,v)) \cdot \vec{N}$$

Take Note :

For an open surface, we can choose the inward/outward normal vector

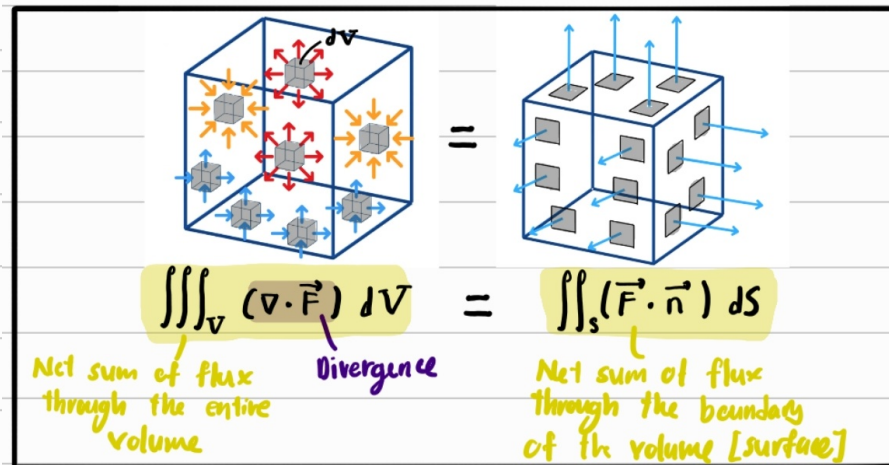
For a closed surface [e.g. sphere], the convention is to choose outward pointing normal. [A +ve flux means the net force is going outwards]

Integral Theorems

$$\iint_S (\vec{F} \cdot \vec{n}) dS = \iiint_V (\nabla \cdot \vec{F}) dV$$

Gauss Divergence Theorem

Total amount of "stuff" being created or destroyed inside a closed space is exactly equal to the net amount of "stuff" flowing out or into its outer surface (shell)



This is due to the fact that each microscopic cube shares walls with its neighbouring microscopic cubes, causing the outward flux of one dV to be the inward flux of the neighbouring dV , thus cancel out.

After cancelling the flux of every single shared internal wall, we are left with flux at the boundary/outer surface of the volume.

Stokes Theorem

$$\oint \vec{F} \cdot d\vec{r} = \iint (\nabla \times \vec{F}) \cdot \vec{n} \, dS$$

This is just Gauss divergence Theorem but instead of divergence, its curl.
And instead of comparing Volume & surface, its surface & line.

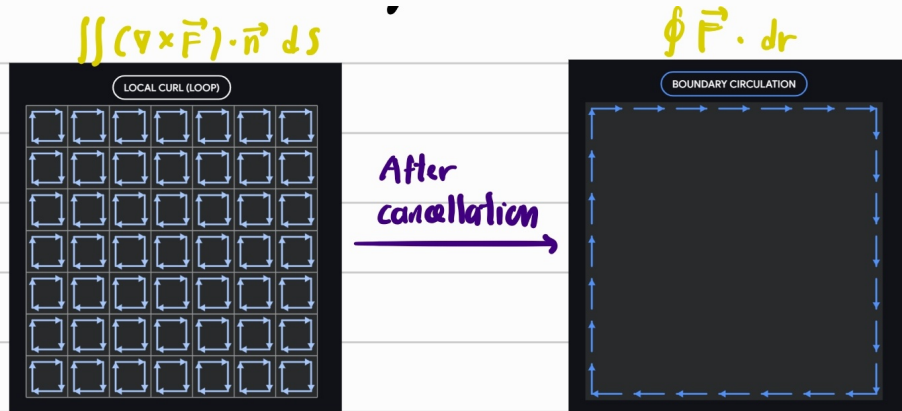
$$\iint (\nabla \times \vec{F}) \cdot \vec{n} \, dS = \oint \vec{F} \cdot d\vec{r}$$

curl
Net sum of curl over the surface

Line Integral
Net sum of curl along the boundary of the surface [Line]

This is due to the fact that each microscopic square shares walls w its neighbouring microscopic square, causing the curl of one dS to cancel out the neighbouring square curl

After cancelling the curl of every single shared internal wall, we are left w curl at the boundary of the surface



Green's Theorem

$$\oint [P(x,y) dx + Q(x,y) dy] = \iint_A \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dA$$

Stokes Theorem but only in xy plane.

$$\text{Stokes Theorem : } \oint \vec{F} \cdot d\vec{r} = \iint_S (\nabla \times \vec{F}) \cdot \vec{n} dS \quad \text{--- ①}$$

$$\text{In xy plane only } \left\{ \begin{array}{l} \vec{F}(x,y) = \langle P(x,y), Q(x,y), 0 \rangle \\ d\vec{r} = \langle dx, dy, 0 \rangle \\ \vec{n} = \langle 0, 0, 1 \rangle \\ dS = dA \end{array} \right.$$

Using eqn ①

$$\oint \begin{bmatrix} P(x,y) \\ Q(x,y) \\ 0 \end{bmatrix} \cdot \begin{bmatrix} dx \\ dy \\ 0 \end{bmatrix} = \iint_S \left\{ \begin{bmatrix} dx/dt \\ dy/dt \\ dz/dt \end{bmatrix} \times \begin{bmatrix} P(x,y) \\ Q(x,y) \\ 0 \end{bmatrix} \right\} \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$\oint [P(x,y) dx + Q(x,y) dy] = \iint_A \left[\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right] dA$$

Laplace Transforms

Laplace Transform transforms a function of real variable (usually time, t) into a function of complex variable (frequency, s)

It's main job is to convert ODE into simple polynomial equations and then convert the solved polynomial eqn back to the time domain to get the result.

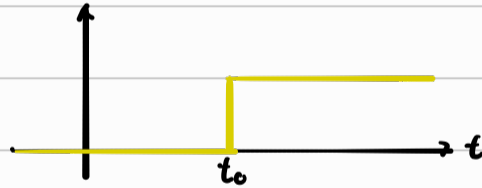
The Laplace Transform of $f(t)$ is

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} f(t) e^{-st} dt$$
$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Unit step function

A function that is either on or off after a certain delay

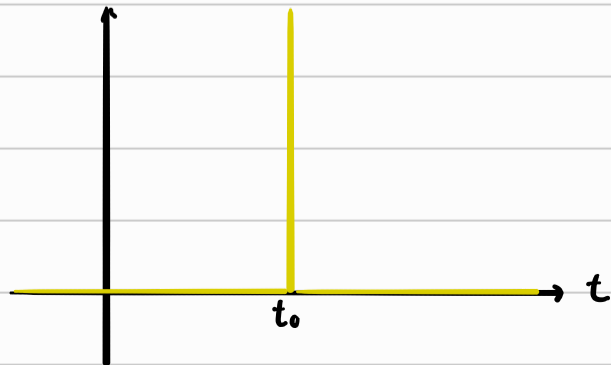
$$u(t - t_0) = \begin{cases} 0 & \text{if } t < t_0 \\ 1 & \text{if } t > t_0 \end{cases}$$



Delta function, $\delta(t)$

A function that simulates an instantaneous pulse

$$\delta(t - t_0) = \begin{cases} \infty & \text{if } t = t_0 \\ 0 & \text{if } t \neq t_0 \end{cases}$$



$$\text{Area of spike} = 1$$
$$\int_{t_0 - \varepsilon}^{t_0 + \varepsilon} \delta(t - t_0) dt = 1$$

Table of Laplace Transforms

$f(t)$	$\mathcal{L}\{f\} = F(s) = \int_0^\infty \left(f(t) e^{-st}\right) dt$
1	$\frac{1}{s}$ for $s > 0$
e^{at}	$\frac{1}{s-a}$ for $s > a$
$\sinh(\lambda t)$	$\frac{\lambda}{s^2 - \lambda^2}$ for $s > \lambda $
$\cosh(\lambda t)$	$\frac{s}{s^2 - \lambda^2}$ for $s > \lambda $
$\sin(\omega t)$	$\frac{\omega}{s^2 + \omega^2}$ for $s > 0$
$\cos(\omega t)$	$\frac{s}{s^2 + \omega^2}$ for $s > 0$
t^n for $n \geq 0$	$\frac{n!}{s^{n+1}}$ for $s > 0$
t^α for $\alpha > -1$	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}}$ for $s > 0$
$g(t) e^{at}$	$G(s - a)$
$tg(t)$	$-\frac{d}{ds}(G(s))$
$t^n g(t)$	$(-1)^n \frac{d^n}{ds^n}(G(s))$
$\frac{dy}{dt}$	$sY(s) - y(0)$
$\frac{d^2y}{dt^2}$	$s^2Y(s) - sy(0) - \frac{dy}{dt}(0)$
$\int_0^t (g(\tau)) d\tau$	$\frac{1}{s} G(s)$
$g(t-a) u(t-a)$	$e^{-as} G(s)$
$\delta(t-a)$	e^{-as} for $a \geq 0$
$(g * h)(t) = \int_0^t (g(\tau) h(t-\tau)) d\tau$	$G(s) H(s)$

Fourier Series

Fourier Series builds on the fact that every graph in the world can be represented by a finite/infinite combination/sum of sine and/or cosine wave(s), with a certain vertical offset.

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right]$$

$L \sim$ half period

y-offset

Shape

Test wave

Test Wave

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$$

$$a_n = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Resonance function

Resonance function

The integrand of a_n and b_n are called resonance function that are made up of multiplying the actual wave, $f(x)$ & the corresponding test wave.

If we were to just integrate the trigonometric test wave across a period, the +ve & -ve area will just cancel out to zero. The only way for the area to not be zero is by multiplying the test wave by a function which is similar to the test wave itself, so that the negative area becomes positive $[-ve \times -ve = +ve]$ and the positive area stays positive $[+ve \times +ve = +ve]$.

As an example, if $f(x) = \cos(x)$, this function perfectly matches the shape of the test wave of a_n , that is also cosine.

Every -ve area will become +ve, every +ve area stays +ve, giving a non-zero total area. $[Integrand = (\cos(x))^2]$

However, for b_n with a test wave of sine, the shape $f(x) = \cos(x)$

is 90° out of sync. While they do sometimes match signs to create +ve area $[-ve \times -ve = +ve]$, they also spend half the period mismatched $[+ve \times -ve = -ve]$.

This creates an equal amount of +ve & -ve areas, thus cancelling out

[This means $f(x) = \cos(x)$ completely has no $\sin(x)$ in it]

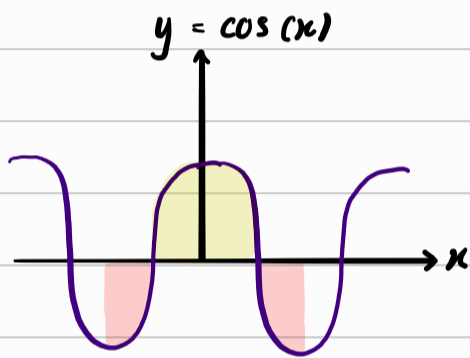
To conclude, if the test wave matches the actual wave, $f(x)$ the integral will yield a non-zero area. So for a_n , the test wave is $\cos(x)$ [Even function], so if $f(x)$ is an even function as well, $a_n \neq 0$, but if $f(x)$ is an anti-even function [Odd function], $a_n = 0$. Same logic applies to b_n with an odd test function, $\sin(x)$.

Recap

Even / Symmetrical function

$$f(x) = f(-x)$$

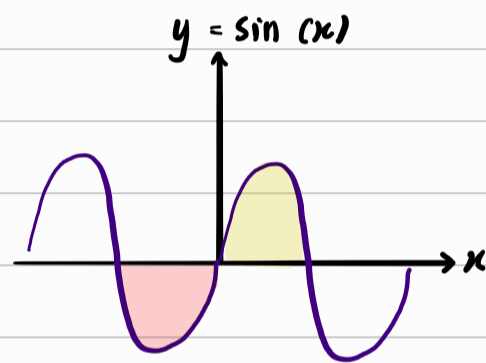
[Mirror about y-axis]



Odd / Anti-symmetrical function

$$f(x) = -[f(-x)]$$

[Mirror about y-axis, then x-axis]



Note that : (Odd func.)(Odd func.) = (Even product function)
 (Odd func.)(Even func.) = (Odd product function)
 (Even func.)(Even func.) = (Even product function)

Rule of thumb for a_n & b_n

$$\int_{\text{Period}} (\text{Odd product function}) = 0$$

$$\int_{\text{Period}} (\text{Even product function}) \neq 0$$

Shortcut for even function

$$\int_{-\pi}^{\pi} (\text{Even func.}) = 2 \int_0^{\pi} (\text{Even func.})$$

Partial Sum, S_N

Fourier Series to a finite number, n

$$S_N = a_0 + \sum_{n=1}^N \left[a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right) \right] \approx f(x)$$

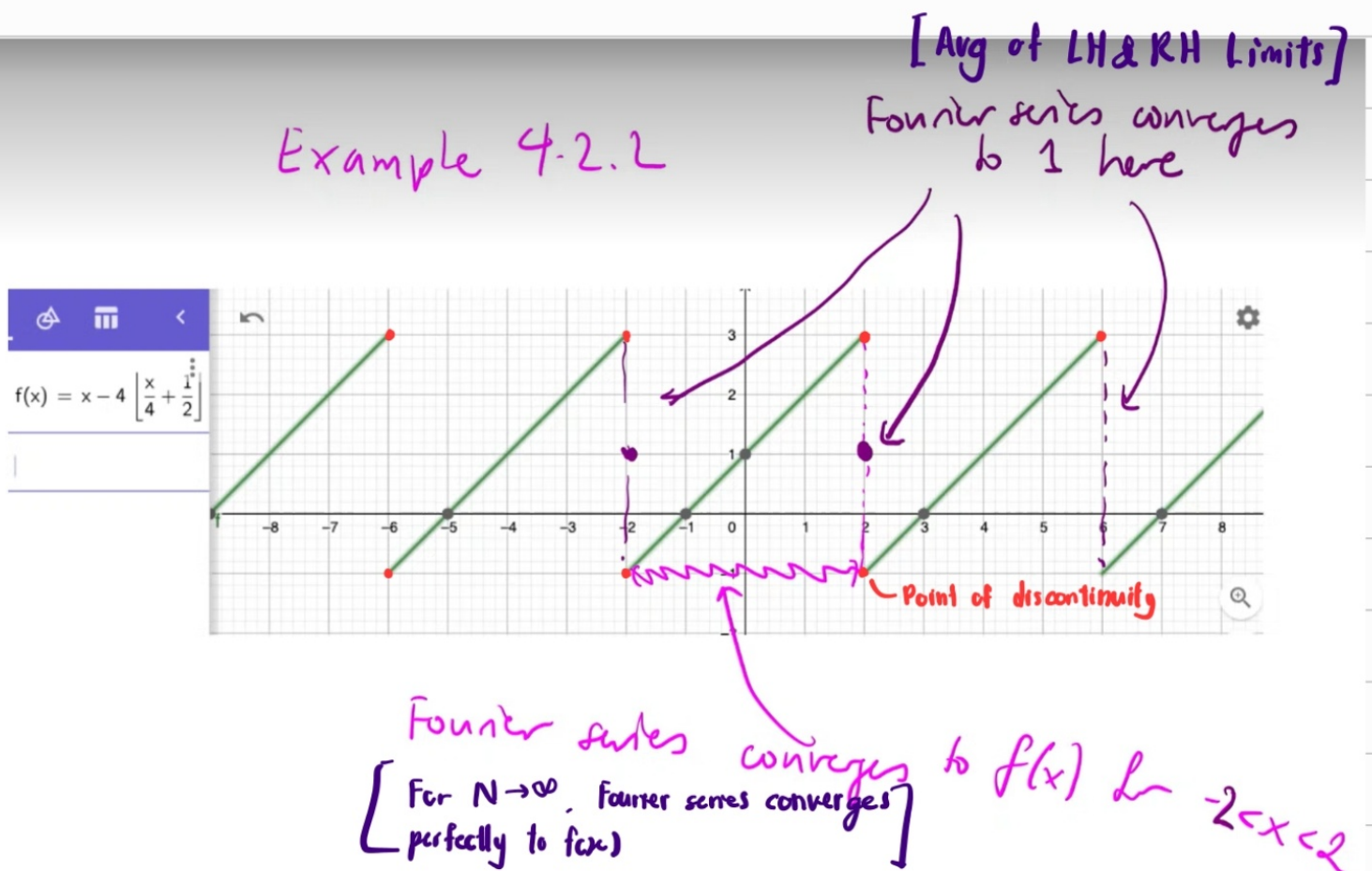
In Fourier Series, when $N \rightarrow \infty$, you will get the exact same graph as $f(x)$. However, to be more realistic we let $N \rightarrow$ finite number.

Thus, S_N will only be an approximation of $f(x)$.

Since S_N is made up of valid functions, its resultant function will also be valid with no vertical jumps.

$\therefore$ It was figured that when $f(x)$ has vertical jumps at x_0 , S_N will converge to the average of LH & RH limits.

$$S_N = \frac{1}{2} \left[f(x_0 - \delta x) + f(x_0 + \delta x) \right]$$



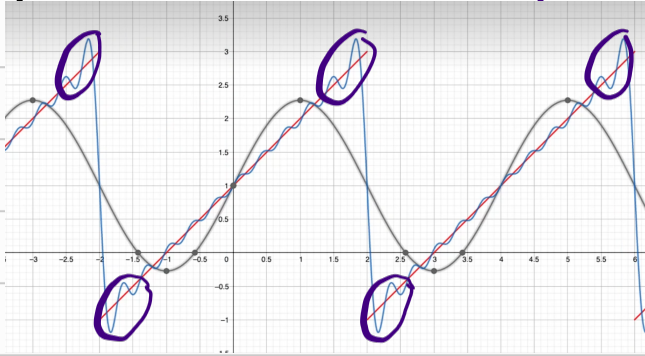
Half Range Expansion Approx.

Mathematical trick to "force" a non-repeating function to work w Fourier Series. We essentially just expand/extend the original function [without changing the original func. within that interval] however we want.

$\therefore$ To make life easier, we expand it to either be $b_n = 0$ perfectly even or $a_n = 0$ perfectly odd.

For the best Fourier Series approx., these are the priorities

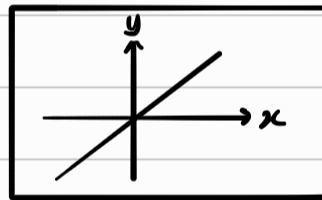
1) Avoid jump discontinuities [Avoid Gibbs phenomenon]



2) Minimise kinks [They reduce the rate of convergence]

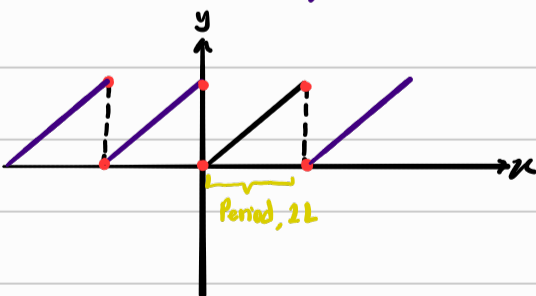
Example

Consider $f(x) = x$, for $0 < x < 1$



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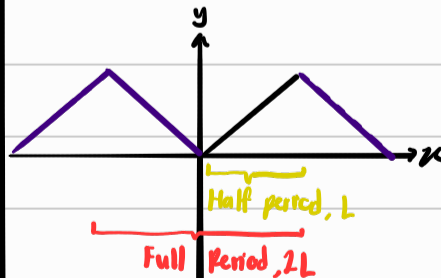
[aka. "Standard/full range expansion"]



Have jump discontinuity

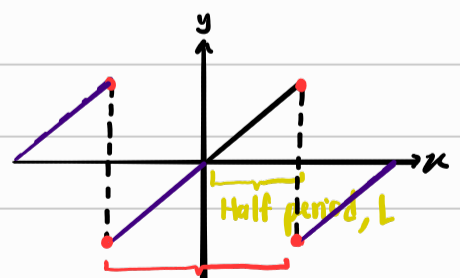
Half - Range Expansion

Even



Don't have jump discontinuity

Odd



Have jump discontinuity

$\therefore$ Even will be used to approx $f(x)$ w Fourier series.

Half-range expansion essentially just gives us the freedom to engineer the boundaries for a better approx w Fourier series

Partial Derivative Eqn (PDE)

one independent variable

ODEs only track how the system changes w respect to one thing at a time [mostly time]

while PDEs takes into acc >1 independent variable at a time

[Normally w respect to time (t) and spacial dimensions (x, y, z) simultaneously.]

General template for Heat/Laplace/Bounded Wave

- ① List down initial & Boundary
- ② Assume separated solution:
 - ↳ Heat/Wave : $u(x, t) = X(x) \cdot T(t)$
 - ↳ Laplace : $u(x, y) = X(x) \cdot Y(y)$
- ③ Substitute into PDE & separate variables
- ④ Set each side equal to a separation const, λ and obtain 2 ODEs
- ⑤ Modify boundary eqn with ② to get "non-trivial boundaries"
- ⑥ Solve for non-trivial solutions for ODE with more "non-trivial boundaries" and get value of λ
- ⑦ Sub λ into other ODE and solve
- ⑧ Sub both non-trivial ODE soln into ②
- ⑨ Use IC/remaining BC to find constants using Fourier Series.

General template for non-bounded Wave

- ① Write D'Alembert's general form : $u(x, t) = F(x+ct) + G(x-ct)$
- ② Solve for F and G using initial conditions given
[Held at rest $\Rightarrow \frac{\partial u}{\partial t} = 0$, shape $\Rightarrow u(x, t)$]
- ③ Sub F & G back into ①

1D Heat Egn

$$\frac{\partial u}{\partial t} = k \nabla^2 u = k \frac{\partial^2 u}{\partial x^2}$$

The rate at which a point heats up or cools down over time ($\frac{\partial u}{\partial t}$) is directly proportional to how much hotter & colder it is than its surroundings ($\frac{\partial^2 u}{\partial x^2}$)

2D Laplace Egn

$$\nabla^2 u = 0 \implies \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

Heat Egn in steady state .

Take note : $\nabla^2 u$ in polar is as follows

$$\nabla^2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2}$$

Advection Equation, 1D

$u(x, t)$

Models pure, rigid transport of physical quantity [e.g. concentration] that is being carried through space with zero diffusion, spreading or smearing involved, thus maintaining its initial shape as it travels.

OR in other words, any local change over time is purely caused by the surrounding flow pushing the spatial shape

Local Rate of change = $\frac{\partial u}{\partial t}$ Ambient Velocity = V Spatial Gradient = $\frac{\partial u}{\partial x}$

$$\frac{\partial u}{\partial t} = -V \frac{\partial u}{\partial x} \quad \text{OR} \quad \frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0$$

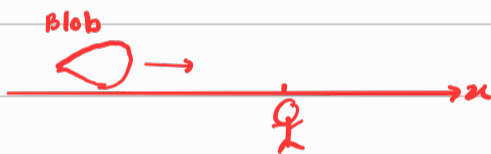
Take note:

Spatial shape $\neq$ Spatial gradient

Why the negative?

Imagine a $+\frac{\partial u}{\partial x}$ [Peak conc. on Right]

Let $V > 0$ [Blob moving right]



From my perspective, the conc. will be decreasing since I'll see the blob from Right to Left [$\therefore \frac{\partial u}{\partial t} < 0$]

Characteristic Curve Derivation

Let $u(x, t) = c$ constant

$$\frac{d}{dt} u(x, t) = \frac{d}{dt} c$$

$$\frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial t} = 0$$

$$\frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \left[-V \frac{\partial u}{\partial x} \right] = 0 \quad \left[\text{using } \frac{\partial u}{\partial t} = -V \frac{\partial u}{\partial x} \right]$$

$$\frac{\partial u}{\partial x} \left[\frac{\partial x}{\partial t} - V \right] = 0$$

$$\frac{\partial u}{\partial x} = 0 \quad \text{or} \quad \boxed{\frac{\partial x}{\partial t} = V} \quad \text{--- } \textcircled{*}$$

[Reject $\therefore$ Trivial]

$\therefore$ This means the only way to guarantee that the conc. around you stays const. [$u(x, t) = c$] is if your personal speed matches the fluid's speed.

This makes sense cuz if $\frac{\partial x}{\partial t} = V$, the relative motion bwn me & the blob is zero. And since the blob spatial shape is constant, the amount of stuff around me will not change

Integrating Eqn $\textcircled{*} \implies \boxed{x(t) = Vt + c}$

current position Starting position
Distance travelled

Since $u(x, t) = \text{constant}$, in other words $u(x, t)$ is only dependent on that constant

$$u(x, t) = f(c)$$

$$\boxed{u(x, t) = f(x - Vt)}$$

This makes sense cuz if my personal speed ($\frac{\partial x}{\partial t}$) matches the Blob's speed (V), our relative motion = 0, thus the conc. ($u(x, t)$) will be constant. The only way to get a diff conc. is by starting at a diff point. And to calc. that starting point, c we just take our [current position, x - Distance travelled, Vt]

Further Explanations

Characteristic Curve $\implies$ General Path of constant concentration

$u(x, t) \implies$ concentration measured at one specific location (x) & time (t)

$f(x)$ or $g(x) \implies$ Reference shape [snapshot]

Connection btwn $u(x, t)$ and $f(x)$

$f(x)$ contains the complete set of every single concentration

reading across all space at one specific slice of time.

$u(x, t)$ simply calculates which specific reading from that

frozen slice ($g(x)$) is currently passing by a location, x at time, t

$$x(t) = c + Vt$$

↓ General version

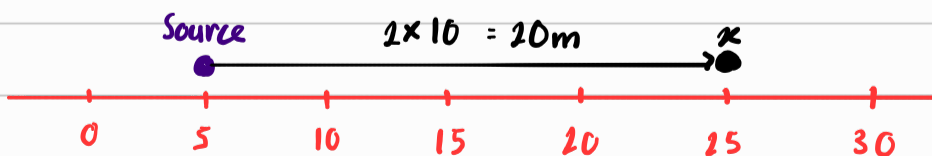
Current position = ^{Snapshot origin / Ref. position} Source Coordinate + Dist. travelled since $f(x)$ snapshot

The corresponding coordinate on reference shape

^{Snapshot origin} Source Coordinate = Current position - Dist. travelled since $f(x)$ snapshot

Example

Let's say $V = 2$, time passed (t) = 10, current position (x) = 25



$$\begin{aligned} \therefore \text{source coordinate} &= 25 - [2 \times 10] \\ &= 5 \end{aligned}$$